Aditya Vikram Singh Dahiya

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https://github.com/AvsDahiya | https://themedimatch.com/

Summary

AI and Machine Learning engineer with a strong foundation in Python and hands-on experience in NLP, supervised learning, and backend systems. Successfully architecting supervised Agentic AI models for multiple services, enhancing customer satisfaction and operational efficiency. Passionate about AI research and building scalable, business-driven workflow solutions, eager to leverage technical depth and creativity to contribute to innovative machine learning projects.

Profile

- **Programming Languages:** Python, Java, C++, Javascript, HTML, CSS, PDDL, R, OCaml, Kotlin, Swift, Ruby
- **Frameworks & Libraries:** TensorFlow, ReactJS, Node.js, API Flask, Firebase, Keras, Scikit-learn, PyTorch, three.js, Flutter, Heroku CLI, AWS S3, Kafka, Google Cloud, Pandas
- **Technical Skills:** MySQL, NoSQL, PostgreSQL, MongoDB, PHPMyAdmin, Git, Microsoft PowerApps, Figma, MATLAB, Scrum & Agile Methods, Relational Databases, Oracle Database, Software Development Lifecycle, DevOps, Full Stack Development
- **Machine Learning & Data Science:** Data Mining, Data Manipulation, Natural Language Processing, LLM Assistants, SLMs, Performance Optimization, Distributed Computing, MLOps, Clustering Algorithms, Computer & Data Visualization, Kubernetes, LangChain
- **Verbal Languages:** English (fluent), Hindi (native)

Education

Heriot-Watt University **2023 - 2026**

BSc (Hons), Computer Science and Artificial Intelligence (GPA: 3.85 expected First Class Hons) *Dubai, UAE*

- **Achievements:** 50% Tuition Scholarship for Academic Excellence, Deputy Principals Award for Academic Excellence
- **Coursework:** AI & Intelligent Agents, Data Mining and ML, Language Processors & Text Analytics, Software Engineering, Web Programming, Data Structures & Algorithms, Operating Systems & Concurrency, Database Management Systems, Cybersecurity & Networks

Jumeirah College **Sep 2014 - Jul 2021**

A-Levels (GPA: 4.0) *Dubai, UAE*

- **Achievements:** A-Levels: A*AB, GCSE: 8A* 2A

Work Experience

Solutions21 FZCO – Expo City Dubai | *Software Engineer* **Oct 2025 - Present**

- Built and extended "Meridian", a multi-tenant PMO SaaS using Next.js 16, React 19, TypeScript, Vercel, Supabase, and PostgreSQL, enabling teams to submit demand approvals, manage projects, create Gantt plans, track timesheets, and view executive KPI dashboards, which streamlined approval processes and improved visibility for executives.
- Implemented tenant-isolated RBAC and workflow architecture, invite-to-access onboarding, auditability, admin tooling, bulk operations, and 79+ SQL migrations to support scalable, secure production use.
- Architected and shipped a 108K-LoC trilingual real-estate marketplace "Iqarat" with 154 API routes, 122 PostgreSQL migrations (30+ tables with row-level security), and a 5-role edge-middleware RBAC system using dual cookie/Bearer authentication for mobile, which launched in three languages and enabled rapid onboarding of new property listings.
- Developed a 9-step property listing engine with bulk-upload, OpenAI-powered NLP search, Google Maps clustering, and an admin console with an 8-KPI dashboard and 10-status × 11-action moderation lifecycle - cutting dashboard load from 10 s to < 3 s.

Virufy.org | *Assoc. Machine Learning Engineer* **Jul 2025 - Sep 2025**

- Trained and optimized CNN, RNN, and transfer-learning models using TensorFlow to detect respiratory diseases from cough and breath audio, achieving a 12% increase in detection accuracy.
- Preprocessed large audio data sets and built scalable ML pipelines on AWS using S3, Lambda, and Step Functions, cutting data-preparat ion time by 40%. Integrated the models into production APIs with Fast API and Docker, establishing automated testing that maintained 99.9% uptime and supported horizontal scaling.

Tech Mahindra Ltd | *AI and Data Analytics Intern* **Aug 2024 - Apr 2025**

- Architected a supervised AI chatbot service using Python and ML frameworks for a banking client, automating account enquiries, fund transfers, and fraud detection while achieving a customer satisfaction rating of over 75%.
- Designed a credit scoring model that utilized a random forest classifier for improved credit risk prediction, reducing false negatives by 20% and automating targeted email marketing.

Alfazance Consulting | *Machine Learning Intern* **Jan 2024 - Feb 2024**

- Engineered an NLP-based AI CV reader using ML frameworks that streamlined candidate selection.
- Integrated the tool into the ERP system, processing over 500 applications weekly and automating data preprocessing pipelines to improve hiring efficiency. Achieved a monthly saving of 30 hours for the HR team by implementing efficient feature engineering techniques.

Expedia Group | *Data Scientist Intern (Remote)* **Aug 2022 - Sep 2022**

- Analyzed KPIs impacting customer interactions using MySQL and PHPMyAdmin for outlier detection, refining data manipulation processes. Built regression-based sorting cost algorithm predicting customer churn with 85% accuracy
- Implemented a k-means clustering algorithm for customer segmentation, enhancing targeted marketing strategies, improving R&D outcomes by 22.5%.

Projects & Publications

From Pre-operative Radiographs to 3D-Printable Hip Implants | *Heriot-Watt Universiy* **Sep 2025 – Apr 2026**

- Undergraduate Hons dissertation project that applies a Pix2Pix conditional GAN to transform de-identified pre-operative AP pelvis radiographs into realistic post-operative total hip arthroplasty (THA) images, using a UNet generator and PatchGAN discriminator trained with adversarial, L1, and SSIM losses.
- Designing a structure-from-motion-informed 2D-to-3D reconstruction pipeline to segment implant geometry from generated radiographs and produce watertight STL models for non-clinical 3D printing, including symmetry-guided femoral stem extrusion and hemispherical acetabular cup fitting.
- Targeting image generation performance of SSIM ≥ 0.80 and reprojection-based reconstruction accuracy of Dice ≥ 0.75, delivering a low-cost, reproducible workflow for radiograph-to-implant planning without additional radiation exposure or patient-identifiable data.

Multimodal Car Analytics for Price Prediction | *Heriot-Watt University* **Sep 2025 – Dec 2026**

- Developed a multimodal used-car analytics pipeline on the DVM-CAR dataset, combining tabular regression models (Linear Regression, Random Forest, and XGBoost) with an EfficientNet-B0 CNN to predict resale price and classify key vehicle attributes from 1,451,784 listing images of 899 UK market car models.
- Delivered held-out test accuracies of 97.44% (manufacturer), 97.21% (body type), 93.84% (model), and 86.39% (colour); the project's strongest resale-price model, Random Forest, achieved R^2 0.966, MAE £1,474, and RMSE 3,683 on 51,205 test samples.

ANN-PSO for Concrete Strength Prediction | *Heriot-Watt University* **Sep 2025 – Dec 2026**

- Developed a predictive modelling pipeline using an Artificial Neural Network (ANN) enhanced with Particle Swarm Optimisation (PSO) to estimate concrete compressive strength from mix-design variables, improving parameter tuning and regression performance.
- Delivered a best model performance of MAE 6.75 ± 0.17 MPa with R^2 0.72 using a wide shallow architecture, with best-test performance reported at approximately 6.7-7.5 MPa MAE.

theMEDimatch.com | *MEDimatch, UK* **Sep 2024 - Present**

- Co-Founder and CTO of MEDimatch, a SaaS platform connecting medical students with research opportunities.
- Building AI-driven matching engine and ERP backend, leading cloud-based infrastructure with multiple users and payment encryption.

H.O.M.E - Heuristical Organisational Monitor of Energy | *Heriot-Watt University* **Sep 2024 - Mar 2025**

- Senior Backend Engineer in a team building a smart home energy monitor application - in accordance with the UAE's COP28 manifesto, adopting Scrum & Agile methodologies to achieve the desired user requirements and deliver the final product on time.
- Developed a scalable API Flask and the Firebase Admin SDK, featuring RESTful endpoints for secure login, registration, and device management with data persistently stored in a Firestore NoSQL database backed by robust bcrypt hashing authentication
- Resolved API bottleneck issues for the high-traffic application, improving response times by 60%, and Diagnosed and resolved system-wide latency issues in a distributed database.
- Integrated a cross platform Flutter frontend that communicates with the Flask backend, deployed on a Heroku CLI cloud server utilising web gunicorn dynos, showcasing a full-stack architecture with real time database interactions and seamless user experience. Performed root cause analysis on application crashes, implementing a fix that achieved 99.5% stability.

Tic Tac Toe Reinforcement Learning AI Agent | *Heriot-Watt University* **Sep 2024 - Dec 2024**

- Architected a modular Java-based Tic-Tac-Toe simulation using a Markov Decision Process (MDP) framework, implementing Value Iteration, Policy Iteration, and Q-Learning agents with state transitions, reward modelling, and environment interactions, supported by a flexible game engine, abstract agent interface, human interaction, and random policy baseline to enable structured comparison of learnt policies against baseline behaviour.

Automated Planning PDDL AI Agent | *Heriot-Watt University* **Sep 2024 - Dec 2024**

- Developed and rigorously tested a suite of PDDL domain and problem files to model an unmanned robotic vehicle (uav), defining environment predicates, action schemas, initial states, goal conditions, validated correctness through planners to ensure optimal solutions and demonstrate understanding of STRIPS-style automated planning.

Supervised Small Language Model Finance AI chatbot | *Tech Mahindra Ltd* **Aug 2024 - Sep 2024**

- Smart Banking Chat Bot—an AI-based project that uses several ML algorithms for natural language understanding to identify intent and entities from user issues (text representation) and generate dialogue using deep learning.

Coleman-Lieu NLP Index Calculator | *GEMS Education* **Dec 2023 - Dec 2023**

- Python-based tool to evaluate textual readability using the Coleman-Lieu Index for pedagogical applications. Enabled targeted text optimisation and facilitated the creation of approved educational reading lists for Kindergarten classroom activities. Actively reduced 20 hours of lesson planning annually, supporting 100+ teachers.

Stock Market Simulator | *Jumeirah College* **Aug 2023 - Sep 2023**

- Flask-based web application simulating real-time stock trading scenarios. Hosted 250+ Jumeirah College Economics Hackathon users. Integrated real stock data APIs and predictive algorithms providing actionable insights, achieving a 40% increase in user engagement.

Awards & Certifications

- **Introduction to Computer Science and Python Programming -** Harvard University CS50 (2023)
- **Nanotechnologist Certification -** Duke University, NC State University, NC Chapel Hill
- **UKMT's Mathematics Challenge Gold + Silver recipient -** United Kingdom Mathematics Trust (2019 - 2021)
- **Gold British Physics Olympiad recipient -** British Physics Olympiad (2019)
- **Physics Student of the Year -** GCSE Academic Excellence (2019)
- **Bronze + Silver recipient -** Duke of Edinburgh International Award (2017 - 2019)
- **Distinction Trinity Music Certifications Plectrum Guitar (grades 4 - 7) -** Trinity College London (2018 - 2020)
- **Bronze Science Crest award recipient -** British Science Association (2016)
- **Diploma in Management and Financial Accounting -** ACCA - Association of Certified Chartered Accountants, UK

Clubs and Societies

Heriot-Watt | *Technical Executive* **Sep 2023 - Jun 2024**

- Google Developer Student Clubs Technical Executive

University of Southampton | *Member* **Sep 2021 - Mar 2023**

- Artificial Intelligence and ECS Society member

SUFST - Southampton Student Formula 1 team | *Lead team engineer* **Sep 2021 - Apr 2023**

- Student Formula 1 team Electronics and Software Department lead team engineer

Jumeirah College | *Head of Innovation & Student Body* **Sep 2015 - Jul 2021**

- Head of Innovation President & Student Council Representative
- Hackathon Team Principal

Charity Work | *Volunteering* **2016 - 2019**

- Gems Swimathon: Fund-raised 2500 AED in sponsorships for building a school in Malawi
- AVT speech therapist @ Tender Hearts Centre for special needs children
- Army Hospital research & referral blood donor | Indian Armed Forces
- Jebel Ali Gurudwara sevadar (40+ hrs)